AASP 2025 Toolbox – README

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Overview

This toolbox is a unified command-line interface that integrates all tasks developed across Assignments 1–5 of the AASP 2025 course. It supports dataset preprocessing, reverse engineering, statistical analysis, anomaly detection, deep learning pipelines, and evaluation workflows for S7Comm-based intrusion detection datasets (QUT and Electra).

Basic Usage

The toolbox is executed via the Python interpreter. Two main flags control execution:

- **-h**: Print help overview and list all available functionalities.
- **-s <keyword>**: Start the toolbox in a specific operation mode.

Example:

```
python toolbox.py -h
```

Operation Modes (-s)

- statistics – Dataset preprocessing and statistical analysis (QUT / Electra).
- reverse_1 – Reverse engineering task 1 (sequence alignment).
- reverse_2 – Reverse engineering task 2 (keyword candidate clustering).
- sda – Stacked Denoising Autoencoder anomaly detection pipeline.
- GAN_raw – GAN-based anomaly detection on raw packets.
- GAN_re – GAN-based anomaly detection on reconstructed packets.
- ngram_detection – N-gram based anomaly detection.
- pipeline – Full Sheet 3–5 pipeline (preprocessing, folds, classifiers, CNN/ResNet, ensemble).

SDA Mode (Stacked Denoising Autoencoders)

The SDA mode supports three sub-modes: model training, feature extraction, and classification based on reconstruction error.

Train models:

```
python toolbox.py -s sda --mode models --pcap-dir-control data/control_pcaps --M 100
--epochs 25
```

Extract features:

```
python toolbox.py -s sda --mode features --pcap-dir-control data/control_pcaps --M 100
--output-dir-features data/features
```

Classify packets:

```
python toolbox.py -s sda --mode classify --pcap-dir-control data/control_pcaps
--pcap-dir-attack data/attack_pcaps --y 0.01
```

Pipeline Mode (Sheets 3–5)

The pipeline mode executes the full experimental workflow including dataset preprocessing, k-fold creation, classical ML classifiers, CNN, ResNet, ensemble classifiers, and plotting.

```
python toolbox.py -s pipeline --mode k_fold --k 5
```

Notes

- Run modes sequentially; files are shared between steps.
- Ensure consistent parameters (e.g., k, M) across dependent modes.
- Use -h to inspect available options for each mode.